

Effect of Running-Induced Fatigue on Tibial Acceleration and the Role of Lower Limb Muscle Strength, Power, and Endurance

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ABSTRACT

DARCH, L., S. CHALMERS, R. CAUSBY, and J. ARNOLD. Effect of Running-Induced Fatigue on Tibial Acceleration and the Role of Lower Limb Muscle Strength, Power, and Endurance. *Med. Sci. Sports Exerc.*, Vol. 55, No. 3, pp. 581–589, 2023. **Background:** High-impact loads have been linked with running injuries. Fatigue has been proposed to increase impact loads, but this relationship has not been rigorously examined, including the associated role of muscle strength, power, and endurance. **Purpose:** This study aimed to investigate the effect of fatigue on impact loading in runners and the role of muscle function in mediating changes in impact loading with fatigue. **Methods:** Twenty-eight trained endurance runners performed a fixed-intensity time to exhaustion test at 85% of $\dot{V}O_{2\max}$. Tibial accelerations were measured using leg-mounted inertial measurement units and sampled every minute until volitional exhaustion. Tests of lower limb muscle strength, power, and endurance included maximal isometric strength (soleus, knee extensors, and knee flexors), single leg hop for distance, and the one leg rise test. Changes in peak tibial acceleration (PTA, g) were compared between time points throughout the run (0%, 25%, 50%, 75%, and 100%). Associations between the change in PTA and lower limb muscle function tests were assessed (Spearman's rho [r_s]). **Results:** PTA increased over the duration of the fatiguing run. Compared with baseline (0%) (mean $\pm$ SD, $9.1g \pm 1.6g$), there was a significant increase at 75% ($9.9g \pm 1.7g$, $P = 0.001$) and 100% ($10.1g \pm 1.8g$, $P < 0.001$), with no change at 25% ($9.6g \pm 1.6g$, $P = 0.142$) or 50% ($9.7g \pm 1.7g$, $P = 0.053$). Relationships between change in PTA and muscle function tests were weak and not statistically significant ($r_s = -0.153$ to 0.142 , all $P > 0.05$). **Conclusions:** Peak axial tibial acceleration increased throughout a fixed-intensity run to exhaustion. The change in PTA was not related to performance in lower limb muscle function tests. **Key Words:** TIBIAL SHOCK, IMPACT LOADING, WEARABLES

Up to 66% of runners will sustain a running-related injury per year, with novice runners at higher risk of developing a running-related injury than experienced runners (1). Bone stress injuries are a common type of running-related injury, which occur as a result of excessive bone loading that overwhelms the bone's capacity for repair (2). An imbalance between the microdamage occurring in bone through loading activities and the formation of new bone can lead to larger scale degradation, potentially leading to the formation of stress injuries (2). With return-to-running time from bone stress injuries taking anywhere between 15 and 31 wk, these can have serious implications for athlete's careers, sport participation,

an individual's mental and physical well-being, and engaging pain-free in activities of daily living (2).

Risk factors for bone stress injuries include female sex, a large increase in training loads, and anatomical factors such as bone density and poor muscle strength (3). Additionally, impact loading-related variables such as high vertical ground reaction force (GRF) loading rates have been proposed to be associated with an increased injury risk when running (4). Given that peak tibial acceleration (PTA) values have been positively and significantly correlated to vertical GRF during running (5,6), it could be assumed that increased PTA may be a risk factor for running-related injury (7). The role of impact loading on injury risk is inconclusive, as the majority of studies identifying a relationship between impact loading and injury risk are retrospective and, therefore, causal links cannot be made. It is important to note that external measurements of load during running are not necessarily representative of damage accumulation at the tissue level (8), yet may provide some insight into the loading experienced during ground contact, and have been shown to discriminate between injured and uninjured groups (9). The incidence of injury over 1 yr is 62% lower when runners undergo gait retraining to reduce their vertical GRF by running "softly" (4). As such, increased PTA and vertical GRF may have a role in the development of bone stress injuries, and a reduction of the rate and magnitude

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of impact loading might reduce the incidence of injuries. Therefore, although the development of bone stress injuries is multifactorial, evidence suggests that measures of impact loading provide useful information in understanding the etiology of this injury in runners.

Running-induced fatigue may play a role on the ability to maintain impact loading, as stresses on the tibia have been shown to increase during a prolonged run (10). Findings regarding the effect of running-induced fatigue on impact loading are also inconsistent, with some studies finding increases (11,12), decreases (13), or no changes to PTA during a run (14,15). Factors that may explain the inconsistency in findings of previous studies include variations in running speed or running surface (13). PTA is highly influenced by speed (16), and a reduction in speed throughout a prolonged run may explain these changes, which would be expected to decrease with slower speeds if all other factors are held constant. Other studies have used fatigue tests of limited duration (<7 min) (17) and tests not being to volitional fatigue (11). Use of a fixed-intensity and exhaustive protocol that controls for these confounding factors would help resolve the uncertainty around the effect of fatigue on PTA and, thus, provide insights into potential injury risk in distance runners.

Muscular contractions are important for attenuating impact forces during running (18). Increases in PTA during running could be explained by peripheral muscle fatigue, which may also compromise the ability to attenuate impact loading through eccentric muscle contractions during early contact, leading to greater impact accelerations during landing (18). A constant paced run at an estimated lactate threshold velocity for 20 min displayed decreased knee flexion and increased PTA over time but no change to whole body shock attenuation (11). This may indicate a shift to a strategy that relies on deformation of passive structures (such as the heel fat pad, tendon, and foot arch compliance) rather than an active mechanism seen through eccentric muscle contractions toward the end of the run (11). When fatigued, adaptation to increased stiffness and increased reliance on passive shock attenuation properties during running may be an indicator of reduced ability of the muscle to attenuate impact loading (2,18). Reliance on more passive shock attenuation methods is related to increased PTA (11,14). Understanding the biomechanical response to fatigue using assessments that are clinically available and simple to perform would allow for assessment for risk of running-related injury without needing the athlete to perform a maximal fatigue test. No studies have investigated if aspects of muscle function (strength, power, and endurance) screened through functional tests are associated with changes to tibial acceleration with fatigue. These tests could suggest aspects of muscle function that may be “protective” for mitigating changes in PTA with running-induced fatigue and play an important role in maintaining segmental acceleration during a fatiguing run.

The aim of this study was to investigate the effect of running-induced fatigue on impact loading in runners and the role of muscle function in mediating changes in impact loading with fatigue. We hypothesized that PTA at foot strike would increase over the

duration of a running-induced fatigue test in comparison with earlier in the run. We also hypothesized that performance in muscle function testing would be negatively correlated to the change in axial PTA with running-induced fatigue (i.e., better strength, power, and endurance would be associated with smaller changes in PTA with fatigue).

METHODS

Participants

To be eligible, participants had to be 18 to 45 yr of age; running at least 20 km·wk⁻¹ for the past month; their most commonly worn running shoes fall within 10%–90% on the Minimalist Index scale (19) (i.e., limiting to more “conventional” shoes because of differences in PTA between different types of footwear [2,16]); and classified as low risk on the Stage 1 Exercise Sport Science Australia Adult Pre-exercise Screening. Participants were excluded if they currently experienced pain during running; experienced a musculoskeletal injury within 4 wk before enrolment (20); had bone, joint, or muscle surgery within the past 12 months; had any medical condition that affects running gait; or ran with a nonrearfoot strike pattern (i.e., the study is restricted to habitual rearfoot strikers because of differences in PTA according to footfall pattern [6]). An *a priori* sample size calculation for a repeated-measures, within-subject design was conducted to determine the required sample size. Assuming a power of 80%, a significance level of 0.05, a small to medium effect size of $f = 0.2$ (based off a study linking greater vertical impact loading rate with increased risk of injury) (21), five measurements, and a correlation between repeated measures of 0.6, 26 participants were required (22). Ethical approval was granted by the University of South Australia’s Human Research Ethics Committee before the commencement of the study (HREC: 203357). All participants provided written informed consent before participating.

Procedures

This study used a within-subjects repeated-measures design. Eligible participants were invited to attend two testing sessions 3 to 7 d apart. Session 1 included recording of participant anthropometrics (age, height, and body mass), muscle function tests (muscle power, strength, and endurance), and an incremental test to exhaustion on a treadmill to determine $\dot{V}O_{2\max}$. Session 2 included a fixed-intensity run to exhaustion at 85% of $\dot{V}O_{2\max}$. This intensity was selected so time to exhaustion could be expected between 20 and 60 min (23). All running measurements were conducted on a calibrated Woodway PPS Med motorized treadmill (Woodway, Waukesha, WI). Participants wore their typical running gear, including their own usual running shoes that met the above criteria for both sessions (19). Participants were asked to avoid strenuous exercise the day before and day of each testing session. Height and body mass were measured (without shoes) using a wall-mounted stadiometer and a set of scales.

All muscle function tests were conducted on both left and right limbs, without shoes. Maximal isometric muscle strength

was assessed using a Mark-10 EK3-500 dynamometer (Mark-10, Corporation, Copiague, NY) recording at 2000 Hz. Handheld dynamometry was chosen over isokinetic dynamometry because of its ease of use, lower cost, and portability, making it a good alternative to expensive isokinetic testing for clinical settings (24). Muscular strength of the knee flexors and extensors and ankle plantarflexors was investigated because of their importance in attenuating impact loading in the early stance phase of running (25). Standardized protocols were used for testing (24,26–28).

Ankle plantarflexors. Ankle plantarflexion strength (focus on soleus) was tested with participants sitting on the edge of a plinth with their foot on the ground, their knee flexed to 110°, and their foot at 75° with respect to the leg. The handheld dynamometer was attached via a nonelastic strap located perpendicular to the foot just proximal to the metatarsal heads with the strap looping around the participants thigh, just proximal to the knee joint. Participants were instructed to plantarflex their foot against the floor to lift their knee into the strap with a 3-s maximal effort. This protocol has been used previously and has excellent interrater reliability (intraclass correlation coefficient [ICC] = 0.99) (28).

Knee extensors and flexors. Knee extensor strength was tested by participants sitting on a treatment table with their hips flexed at 90° and tested knee flexed at 30° (anatomical reference position is 0° with knee in extension). The handheld dynamometer was fixed to the floor using a nonelastic strap with the strap being positioned perpendicular to the end of the tibia, just proximal to the malleoli. Participants grasped the edge of the table for stabilization and attempted to extend their knee with a 3-s maximal effort. This methodology has excellent interrater reliability (ICC = 0.96) (26). Knee flexor strength was tested with participants lying prone on a treatment table and their ankle hanging off the bed. The hip was positioned in the anatomical reference position (0°) with the knee fully extended (0°). Padding was placed just above the underside of the knee for comfort. The handheld dynamometer was attached to a nonelastic strap positioned perpendicular to the participants tibia just proximal to the malleoli and attached to the floor beneath the participant. The participants grasped the edge of the table for stabilization and attempted to flex their knee with a 3-s maximal effort. This protocol has excellent interrater reliability (ICC = 0.84) (24). For all tests, participants were given two practice attempts, with three trial attempts, with 30-s rest between attempts (24). If the final value test was the largest, participants continued until they were unable to improve. The highest value was used for analysis (24), apart from where the largest value was more than 20% larger than the mean of attempts when an average of the trials was used for analysis instead (27). Verbal encouragement was given for all tests to elicit maximum effort. Reliability testing for the muscle function tests was conducted on 10 healthy individuals before the main study on the same day with a 10-min rest between trials, and results can be found Supplemental Table 3 (see Supplemental Digital Content, <http://links.lww.com/MSS/C727>).

TABLE 1. Characteristics of study participants.

	Total	Female	Male
Sample size (n)	28	13	15
Age (yr)	38.4 ± 5.5	38.0 ± 6.0	38.8 ± 5.3
Body height (m)	1.7 ± 0.1	1.7 ± 0.1	1.8 ± 0.1
Body mass (kg)	71.0 ± 11.7	62.5 ± 8.2	78.4 ± 9.0
BMI (kg·m ⁻²)	23.6 ± 2.7	22.9 ± 3.1	24.1 ± 2.3
VO _{2max} (mL·kg ⁻¹ ·min ⁻¹)	52.4 ± 7.9	47.1 ± 6.5	57.1 ± 6.1
Running economy (mL·kg ⁻¹ ·km ⁻¹)	197.5 ± 15.0	191.5 ± 11.4	201.6 ± 16.2
Speed at 85% of VO _{2max} (km·h ⁻¹)	13.6 ± 1.8	12.5 ± 1.6	14.6 ± 1.5
Time to exhaustion (min)	33.8 ± 14.5	34.2 ± 14.7	33.5 ± 14.9

All data are presented as mean ± SD.

Muscular power was assessed using a single leg hop for distance test (29). Participants were instructed to stand on one foot and to hop as far forward as possible while landing on the same foot while maintaining balance. Using upper body momentum was allowed. Participants conducted a self-selected warm-up, which involved body weight squats and lunges. Participants were given one practice trial, followed by three hops, with an additional hop given if they lost balance on any attempt. This method has good test–retest reliability (ICC = 0.96) (29). A 30-s rest was given between each trial (29). The maximum hop distance was measured from the front of the foot before hopping to the heel of the same foot after landing.

Muscular endurance was assessed using a one leg rise test (30). Participants repeatedly rose on one leg from a sitting to standing position (and returned to sitting) from an adjustable height plinth while exhibiting full postural control without the use of upper body momentum (determined by observation of upper limb movement). The plinth height was set so the sitting position resulted in a knee angle of 90° with the foot positioned flat on the ground (30). The number of successfully completed repetitions (concentric and eccentric phase) was counted. The test ceased when the participant was unable to perform another rise (30). This test was selected because of its knee extensor endurance requirements, which may play a role in knee-related changes due to fatigue in runners (17).

Maximal oxygen uptake was assessed via a calibrated Parvo metabolic cart (Parvo Medics, Salt Lake City, UT) during an incremental test to exhaustion on a calibrated treadmill after a 30-min break from the final muscle function test. The treadmill speed started at either 8 or 10 km·h⁻¹ (depending on estimated aerobic capacity) for the first 4 min, where running economy was calculated (31). Running economy was calculated as the average oxygen uptake during the final 2 min, divided by running speed in kilometers per minute (mL·kg⁻¹·km⁻¹) (31). Immediately after this stage, the treadmill speed was increased by 1 km·h⁻¹·min⁻¹ for 5 min, where after the treadmill speed was increased by 0.5 km·h⁻¹ until volitional exhaustion of the participant. VO_{2max} was calculated as the average of the two largest consecutive values recorded for oxygen uptake (mL·kg⁻¹·min⁻¹), where sampling period was 15 s. Heart rate and RPE (6–20) (32) were measured at the end of every minute. Heart rate was measured using a Polar chest strap heart rate monitor (RS800CX; Polar Electro Oy, Kempele, Finland) and recorded manually as the value at the end of each minute. PTA was recorded during an initial 4-min stage at a fixed running speed, before the start

of the 1-min speed increments. A 20-s sagittal-plane video (30 Hz) using an iPad (Apple, Cupertino, CA) during the initial 4 min was recorded to assess participants' foot strike pattern. The iPad was located on the right side of the participant set at the height of the foot.

Session 2 involved a fixed-intensity run to exhaustion at 85% of $\dot{V}O_{2\max}$. Participants were given a 6-min warm-up at either 8 or 10 $\text{km}\cdot\text{h}^{-1}$ on the treadmill running (33). From here, participants continued to run at a speed associated with 85% of their $\dot{V}O_{2\max}$ speed until volitional exhaustion. A fixed-intensity protocol removes the influence of changing speed on PTA (16). Assessing PTA across multiple time points during the run allows us to assess where the most amount of change in PTA occurs. Treadmill gradient remained at 1% throughout both runs (34). Heart rate, RPE (6–20) (32) and rating of fatigue (ROF) (0–10) (35) was recorded every 3 min. Participants were permitted water *ad libitum*; however, no food or electrolyte/carbohydrate drinks were allowed. Participants had two fans directed at them and were verbally encouraged throughout.

Tibial accelerations were recorded using a triaxial accelerometer (Blue Trident IMU; Vicon Motion Systems Ltd., Oxford, United Kingdom; dimensions = $42 \times 27 \times 11$ mm, weight = 12 g). Tibial accelerations provide a measure of impact loading, and peak axial tibial acceleration is strongly correlated with average and instantaneous GRF load rates during running (5,6). Inertial measurement unit (IMU) were attached to the left and right distal anteromedial shank, just superior to the medial malleolus, using the provided strap tightened as comfortably possible and then covered with additional wrapping (Superwrap, FabriFoam, Exton, PA) to reduce additional movement (16,36). The IMU was positioned so the y-axis aligned with the long axis of the tibia and data were sampled at 1125 Hz (16). PTA was measured for approximately 25 s at the end of each minute throughout the fatiguing run. PTA was also measured in the warm-up stages during both sessions to measure intersession reliability (see Fig. S5, Supplemental Digital Content, Tibial acceleration measured during both warmups, <http://links.lww.com/MSS/C727>). The same IMUs were used on each leg across testing sessions and participants.

Data Processing and Analysis

Accelerometer data were downloaded in Capture.U (v.1.3.0; Vicon Motion Systems Ltd., Oxford, United Kingdom), and a custom MATLAB (2020b; MathWorks, Natick, MA) script was used to analyze the data. Sampling theory states that the sampling rate should be at least twice the highest natural frequency content of the signal (16). A power spectral analysis of data during running shows that 99% of a signals power is contained below 60 Hz (37). As such, the y-axis (axial) acceleration data were downloaded at 500 Hz, which adequately accounts for any higher frequency noise. The original data file was reduced to the middle 20 s of each sampling period, corresponding to approximately 25–30 foot strikes based on participant running speeds of 9.9–16.9 $\text{km}\cdot\text{h}^{-1}$. Data were normalized to gravitational equivalents ($g = 9.81 \text{ m}\cdot\text{s}^{-2}$). Data were filtered

using a second-order, low-pass Butterworth filter with a cutoff frequency of 60 Hz (37). Peak axial accelerations were determined as the peak positive acceleration from each stance phase and averaged to provide a mean PTA. Data points that were more than three standard deviations away from the median of that time point were removed. This process was automatically repeated for each time point recorded. Where there were single peaks that exceeded the low-g accelerometers capability ($>15.9g$), the single peak was removed from the analysis. If a participant had consistent peaks above the capability for the accelerometer, data from the other limb were used (occurred for four participants). If both limbs had data consistently exceeding the capability of the accelerometer, that participant was excluded from subsequent analysis (occurred for one participant). A custom Excel spreadsheet was then used to determine the data point that was closest to each time point of interest (1st, 25%, 50%, 75%, and final/100%, hereby referred to as 0%, 25%, 50%, 75%, 100%, respectively).

Statistical Analysis

All testing was conducted on both legs, but only data from one leg (selected randomly by a computer) were used for analysis to maintain independence of data (38). Reliability of maximal isometric muscle strength and PTA was determined using an ICC (two-way mixed model with absolute agreement) with 95% confidence intervals, coefficient of variation (CV), and minimal detectable change (MDC_{95}), which can be found in Supplemental Table 3 (see Table S3, Supplemental Digital Content, Reliability scores for muscle function tests and peak tibial acceleration, <http://links.lww.com/MSS/C727>).

Friedman's tests were used to investigate differences in means of ROF, RPE, heart rate (percent of maximum), and PTA (absolute and expressed as change from 0%) at different time points. Friedman's test was used as these data sets contained outliers as determined by visual inspection of box plots. Where significant main effects were detected, *post hoc* tests with a Bonferroni adjustment were conducted. Descriptive data for Friedman's test are presented as median (interquartile range = 25%–75%). Effect sizes with 95% confidence intervals were presented between each time point (25%, 50%, 75%, and 100%) with 0% as the reference. Effect sizes (r) were calculated as recommended by Field (39). Spearman's rank-order correlation coefficients (r_s) were used to determine the direction and magnitude of the relationship between the muscle function tests (strength, power, and endurance) and the change in axial PTA during the fatiguing run. The PTA change value was the maximal value obtained at any time point minus the reference (0%) value. Spearman's correlation was used as not all data were normally distributed (knee extensor, Shapiro–Wilk, $P = 0.004$). Partial eta squared (η^2) effect sizes are reported descriptively to describe the effect of gender on PTA. Effect sizes were classified as small (0.009), medium (0.059), and large (0.138) (40). Further exploratory analyses were conducted to explore the correlation between aerobic capacity (relative $\dot{V}O_{2\max}$), running economy, and change in axial PTA. The level of significance

was set at $\alpha = 0.05$. Statistical analysis was conducted using SPSS (v. 26; IBM, Armonk, NY).

RESULTS

Participants. Forty-four participants expressed interest, and 34 agreed to participate in this study. Five participants dropped out of the study after the first session due to injuries, illness, or logistics associated with the COVID-19 pandemic. One participant was removed from the analysis because of PTA data exceeding the capability of the accelerometer on both legs. Therefore, 28 participants (13 females, 15 males) are included in the analysis. Participant characteristics can be found in Table 1.

Reliability tests. The absolute mean difference between warm-ups before the $\dot{V}O_{2\max}$ (session 1) and the fatiguing run (session 2) for PTA was 0.514g (0.445). Intersession reliability was good (ICC = 0.839, 95% CI = 0.625–0.931, CV = 0.05). Intratester reliability for the muscle function tests ranged from 0.82 to 0.89, CV = 0.110–0.143 (see Table S3, Supplemental Digital Content, Reliability scores for muscle function tests and peak tibial acceleration, <http://links.lww.com/MSS/C727>).

Fatigue measures. RPE and ROF at the end of the run were significantly different relative to 0% (main effect $P < 0.001$). Compared with the start of the run, RPE, ROF, and heart rate increased at 50%, 75%, and 100% of run duration ($P \leq 0.001$ for all, effect size = 0.353 to 0.777) (Table 2 and Figs. S2–S4, Supplemental Digital Content, Rating of perceived exertion, rating of fatigue, and heart rate at each time point during the fatiguing run, <http://links.lww.com/MSS/C727>). When compared with 25%, RPE, ROF, and heart rate increased at 75% and 100% ($P < 0.001$ for all, effect size = 0.404 to 0.619). When compared with 50%, RPE, ROF, and heart rate increased at 100% ($P < 0.001$ for all, effect size = 0.384 to 0.400). When compared with 75%, ROF increased at 100% ($P = 0.032$, effect size = 0.192) (see Table S1, Supplemental Digital Content, Rating of perceived exertion, rating of fatigue, and heart rate response comparisons between timepoints throughout the fatiguing run, <http://links.lww.com/MSS/C727>).

Peak axial tibial acceleration. Absolute and change scores for PTA are presented in Figure 1 and Table 2 (see Table S2, Supplemental Digital Content, Peak tibial acceleration [absolute values and changes] comparisons between timepoints throughout the fatiguing run, <http://links.lww.com/MSS/C727>; and Figure S1, Peak tibial acceleration expressed as change from baseline at each time point during the fatiguing run, <http://links.lww.com/MSS/C727>). Absolute PTA and relative change in PTA increased significantly throughout the run (main effect $P < 0.001$ for both). Compared with 0%, changes in PTA were evident after 75% and 100% of the run duration ($P = 0.001$ and <0.001 , with effect sizes of 0.321 and 0.450, respectively). When compared with 25%, absolute and relative changes in PTA were evident at 100% ($P = 0.041$, effect size 0.243). When compared with 0%, increases in PTA values at 25% and 50% of the run duration were not different ($P = 0.142$ and $P = 0.053$, with effect sizes of

TABLE 2. RPE, ROF, heart rate responses, and PTA (absolute values and changes) throughout the fatiguing run.

Time	0%	25%	50%	75%	100%
RPE (6–20)	10.5 (8.7–11.0)	13.0 (12.0–14.0), $P = 0.794$, $r = 0.154$	15.0 (14.1–16.0), $P < 0.001$, $r = 0.388$	17.0 (16.0–18.0), $P < 0.001$, $r = 0.569$	19.0 (18.0–19.0), $P < 0.001$, $r = 0.773$
ROF (0–10)	3.0 (2.0–3.5)	4.0 (3.0–5.0), $P = 0.736$, $r = 0.157$	6.0 (5.0–7.0), $P = 0.001$, $r = 0.353$	7.0 (7.0–8.0), $P < 0.001$, $r = 0.553$	9.0 (9.0–9.5), $P < 0.001$, $r = 0.741$
HR (% of maximum)	76.2 (70.7–80.7)	89.1 (86.5–91.1), $P = 0.655$, $r = 0.162$	92.9 (90.9–95.9), $P < 0.001$, $r = 0.377$	94.5 (93.3–97.4), $P < 0.001$, $r = 0.569$	97.4 (95.2–100), $P < 0.001$, $r = 0.777$
Absolute PTA (g)	8.9 (8.0–10.0)	9.8 (8.6–10.7), $P = 0.142$, $r = 0.207$	10.0 (8.6–10.7), $P = 0.053$, $r = 0.236$	9.8 (8.8–10.9), $P = 0.001$, $r = 0.321$	10.0 (9.2–11.3), $P < 0.001$, $r = 0.450$
PTA change (g)	–	0.46 (0.05–0.85)	0.48 (0.01–1.03)	0.62 (0.35–1.30)	0.88 (0.42–1.49)

Data are presented as median (25–75 percentile). HR, heart rate; r , effect size compared with 0%.

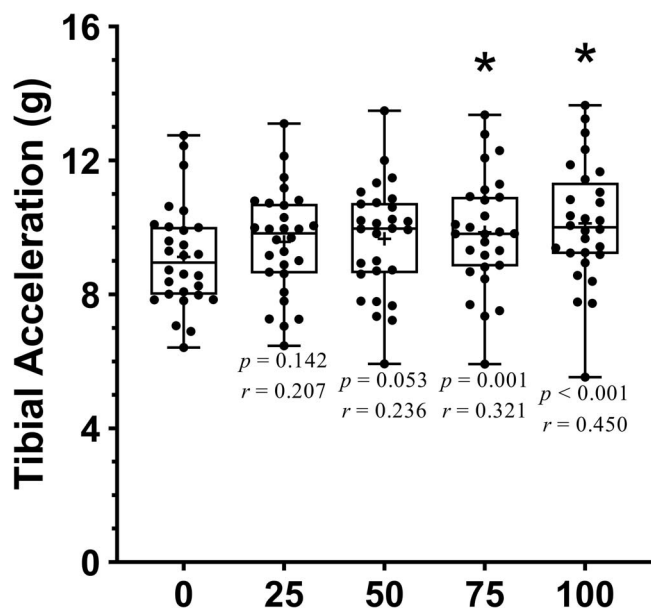


FIGURE 1—Absolute PTA at each time point during the fatiguing run. Whisker plot limits show min to max. Outsides of the box represent the first and the third quartiles. The middle line represents the median, and the plus (+) symbol represents the mean. Gray dots represent individual participants. The asterisk (*) represents statistically significant ($P < 0.05$) increases from the first time point. r , effect size compared with 0%.

0.207 and 0.236, respectively). When compared with 25%, PTA at 50% and 75% were not different ($P = 1.000$ for both, effect size 0.029 to 0.114). When compared with 50%, PTA values at 75% and 100% were not different ($P > 0.112$ for both, effect size = 0.086 to 0.214). When compared with 75%, PTA at 100% was not different ($P = 1.000$, effect size = 0.129).

Differences in PTA between gender. Differences in PTA between males and females displayed small effect sizes (partial $\eta^2 = 0.001$ and 0.033, respectively) (see Supplemental Fig. S6, Supplemental Digital Content, Absolute peak tibial acceleration at each time point during the fatiguing run split by sex, and Figure S7, Peak tibial acceleration measured as change from baseline at each time point during the fatiguing run split by sex, <http://links.lww.com/MSS/C727>).

Associations between muscle strength, power, and endurance and change in PTA. Knee extensor and flexor strength results were not different between left and right legs as determined by a paired-samples t -test ($P = 0.160$ – 0.560); therefore, ICC values are provided using both legs. Data for soleus were statistically different between sides as determined by a paired-samples t -test and, therefore, are provided with both limbs and combined scores ($P = 0.560$ and 0.002 for first and second measurements, respectively). Correlations between PTA change value and muscle function tests were weak and not statistically significant (Table 3). Similarly, there was no statistically

significant correlation between PTA change value and performance tests ($\dot{V}O_{2\max}$ $r_s = 0.071$, $P = 0.719$, running economy $r_s = 0.299$, $P = 0.146$; Table 3). Descriptive data for the muscle function tests can be found in Supplemental Table S4 (see Table S4, Supplemental Digital Content, Measure of central tendency and variance for muscle function tests, <http://links.lww.com/MSS/C727>).

DISCUSSION

We hypothesized that PTA would increase over the duration of an exhaustive fixed-intensity run and that the degree of change would be negatively correlated to performance in selected lower limb muscle function tests. The first hypothesis was partially supported, with PTA increasing from 75% of the run duration. However, the second hypothesis was not supported, as correlations between change in PTA and lower limb muscle function tests (strength, power, and endurance) were weak and not significant.

This study is the first to investigate if aspects of muscle function screened through functional tests are associated with changes to PTA with fatigue. To our knowledge, only one other study has investigated PTA changes over the duration of an exhaustive fixed-intensity run (12). In support of our first hypothesis, PTA at foot strike increased from 75% of a running-induced fatigue test in comparison with the start of the run. Although research investigating impact loading and injury risk uses GRF, PTA is correlated to vertical GRF and load rates (5,6). As such, some evidence links increased PTA with increased risk of injury in runners (4,21). This should be considered with caution, as the development of injuries are complex, and inferring a modified risk of injury from a single biomechanical metric (such as PTA) fails to consider the multifactorial nature of injuries and lack of knowledge about internal bone loading conditions (41). Edwards (41) postulates that the relationship between loading rate and injury risk is real, but that increased loading rates are also associated with changes to other variables that also increase tissue specific loads (stress magnitude). PTA increased from 75% of the run duration, highlighting that PTA is elevated before runners reach exhaustion; however, substantial fatigue and exhaustion may be required before these changes are observed.

Perceived exertion increased (50% of run completion) before increases in PTA (75% of run completion). Camello et al. (15) investigated PTA during a submaximal run, with participants reporting a fatigue scale 5 out of a possible 10 using a visual analog scale rating at the end of the run, finding no changes to PTA. A fixed 30-min run was used in this study, which is similar to the mean duration of our study. This highlights that fatigue, rather than time, may be the important factor in regulating PTA. Using a protocol that resulted in greater self-reported fatigue (mean

TABLE 3. Correlations between change in PTA, muscle function tests, and performance metrics.

		MIMS (Soleus)	MIMS (Knee Extensor)	MIMS (Knee Flexor)	SLHD	OLRT	$\dot{V}O_{2\max}$	RE
Change in PTA (g)	r_s	-0.036	0.109	-0.090	0.060	0.088	0.071	0.299
	P	0.857	0.582	0.649	0.762	0.657	0.719	0.146

SLHD, single leg hop for distance; MIMS, maximal isometric muscle strength; OLRT, one leg rise test; $\dot{V}O_{2\max}$, maximal oxygen consumption; RE, running economy.

RPE 6–20 = 17.7), Clansy et al. (18) had participants finishing their run at a perceived exertion levels similar to what we observed at our 75% run duration (RPE 6–20 = 17.0); however, no changes in PTA was observed. This could be explained by the protocol used, where participants completed two 20-min running bouts rather than a single continuous run. Derrick et al. (12) observed increased PTA during an exhaustive run but did not provide measures of perceived exertion throughout the protocol. The exhaustive run protocol, by nature, implies that participants should be substantially fatigued. On average, participants ran for 15.7 min, again suggesting that fatigue rather than time is more important for mediating changes to PTA. By measuring perceived exertion throughout the exhaustive run, we can show that increased levels of exertion are required to elicit changes to PTA. This has important implications for the ecological validity of the work. During a typical prolonged submaximal training run, it is likely that runners would regulate their speed (and therefore an altered PTA) before reaching these high levels of fatigue. Any number of gait-related changes could explain the observed differences to PTA, such as changes to stride length, leg stiffness, shock attenuation, kinematics, or muscle activation.

We observed an increase in PTA of about 1g over the duration of the run, from a start point of 8.9g. Gait-retraining interventions are able to reduce PTA, with Clansy et al. (42) maintaining ~2g (31%) and Crowell and Davis (43) ~3.5g (44%) decrease in PTA 1 month postintervention after six and eight sessions of real-time feedback gait retraining. The importance of a 1g increase in PTA on injury risk is unknown, as studies identifying risk of injury from impact loading (21) or using gait retraining to reduce injury occurrence (4) have used vertical GRF. More research is needed to know the importance of a 1g increase in PTA on injury risk. Minimal detectable change (MDC_{95}) for PTA in our study was 0.77g, giving confidence that the group-level change observed with fatigue is above measurement error.

Refuting our second hypothesis, change in PTA was not correlated to performance in the selected muscle function tests. The premise underlying this hypothesis was driven by two factors: surveys of training habits in distance runners indicate that resistance training is commonly performed for injury prevention (44), and evidence from randomized control trials indicate that strength training assists in the prevention of sporting injuries (45). The effects of isolated muscle fatigue on PTA have not been assessed during running. The use of tibia modeling shows no changes to tibia stresses before and after a fatiguing calf-raise protocol during running (46); however, tibia stresses increase during prolonged running (10). As such, tibial stresses appear unrelated to isolated plantarflexor muscle fatigue, highlighting the complexity between fatigue and lower limb loading. Human pendulum protocols, where the foot is impacted into a force platform with the person lying supine on a swing, show localized fatigue to the tibialis anterior and gastrocnemius result in reductions in PTA, likely caused by the reduced force generating capacity of the muscle (16). How this relates to running is currently unknown, but it is likely that localized fatigue to muscles would impact muscle coactivation, the distribution of work between muscle groups, and play a role in changing PTA (47).

On average, PTA occurs within the first 30 ms of the gait cycle from initial contact (37). Given that the tests used to assess muscle function are not specific to rapid eccentric contractions, it is possible that other features of muscle function may provide more insight into if and how the body can effectively attenuate impact loading. Investigating preactivation of the muscles involved in the eccentric component of the loading response is one line of investigation to further understand the muscular control of impact loading and changes with fatigue.

Limitations. This study externally fixed intensity so that participants did not alter their running speed as a consequence of fatigue. This has implications for real-world translation of the results, as even with set pacing, it is unlikely runners will run at the same speed for an entire competitive event. Despite this, the current protocol design allows the isolation of the effect of fatigue on PTA. Although treadmill and overground running is comparable (48), PTA is lower when measured in a laboratory rather than outdoor or in-field setting (13). Although this is unlikely to impact the change scores as the surface was held constant, the absolute values for PTA will be different to overground or in-field-based testing, and as such, comparisons between this field-based literature should be done with caution. Another limitation is using PTA to infer risk of injury. Currently, research that links injury risk with greater impact loading uses measures of GRF, not PTA (7,21). Despite this, PTA is correlated with vertical GRF and load rates (5,6), suggesting that increases in PTA could impact risk of injury.

Because of complications with the high-g sensor, this study used the low-g sensor with a limit of 16g. Any signal above this was clipped and unable to be recorded. However, only one participant was removed from the analysis because of PTA data exceeding the capability of the accelerometer on both legs.

Sex differences were investigated as part of an exploratory analysis; as such, the current study and sample size were not powered for this analysis. Future studies specifically investigating sex-related differences would be a valuable addition given differences in injury risk. Further, no significant correlations between predictors of running performance (running economy at 10 km·h⁻¹ and $\dot{V}O_{2max}$) and change in PTA were observed. As such, our evidence does not support the notion that PTA is able to be more regulated by runners of differing sex or higher endurance performance (as indicated by greater $\dot{V}O_{2max}$). Competitive runners have less change in kinematics than novice runners during a fatiguing run (49); however, under negligible fatigue, running kinematics at a preferred speed are comparable (50). Running economy was calculated at a fixed 10 km·h⁻¹ and, as such, may not be an ideal measurement for runners who habitually train at faster speeds than this. Finally, we did not take into consideration the cumulative running load between first and second sessions, so fatigue accumulated before the sessions may have impacted results.

CONCLUSIONS

In conclusion, this study has identified that PTA increases over a fixed-intensity run to volitional exhaustion. Importantly,

changes in PTA were observed from 75% during the run at high levels of perceived exertion and fatigue, suggesting that increases in impact loading may be of practical significance to distance runners before volitional exhaustion. The change in PTA was not explained by performance in selected tests of lower limb muscle strength, power, and endurance. These findings highlight that some commonly used clinical tests may not be specific enough to infer which runners would increase impact loading during a fatiguing run. As PTA occurs early in loading response, future research should investigate the relationship between muscle preactivation and PTA.

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Participants signed informed consent forms before involvement in any testing. Participants were aware that the study was conducted with the purpose to be published.

The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine.

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